Energetic Particle Detector Review Update

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March 28, 2026

Orbit-Based View of Detector Developments

- Instruments are grouped by measurement environment:
  - heritage LEO mapping / dosimetry systems
  - MEO radiation-belt instruments
  - cislunar / lunar-environment concepts
- Highlighting used in the tables:
  - Red: not in the original baseline list
  - Blue: baseline instrument with updated / changed performance or status

LEO Instruments

Instrument	Type	Proton (MeV)	Mass	Status
HMRM	Pixel Si	0.6–500	52 g	flown
LUCID	Timepix array	>10	1.2 kg	flown
SATRAM	Timepix	ND	172–380 g	flown
MIRAM	Timepix3 + Si	10–400	>150 g	proposed / development
MERiT	SSD + APD	7–100	1.13 kg	flown
RADMON (Aalto-1)	Si + CsI	10–80	360 g	flown
RISEPix	2× Timepix stack	ND	>1 kg	flown
VZLUSAT-1	Timepix	ND	ND	flown
ALTEA	Si telescopes	ND	ND	flown
DosTel	Si telescope	ND	ND	flown
TEPC	Gas proportional counter	ND	ND	flown
HEPD	Si + scintillator	30–200	43 kg	flown

LEO remains dominated by miniaturised, mapping, and dosimetry-oriented systems, especially Timepix-class detectors.

MEO Instruments

Instrument	Type	Proton (MeV)	Mass	Status
EMU	Si telescope	15–100	3.5 kg	flown
Merlin	Si + SURF	>40	1.7 kg	flown
CEDEX	Si-based	ND	ND	flown
SREM (GIOVE-B)	Multi-detector	23–450	2.6 kg	flown
NGRM	Si (PSI)	2–200	1.4 kg	flown
ICARE-NG	Si coincidence	27–292	2.8 kg	flown
CEASE3	Si stacks	2–150	4 kg	flown
CREDANCE	Si detectors	ND	ND	flown
REPT	Si stack	17–200	13.4 kg	flown

MEO systems are primarily radiation-belt instruments with thicker silicon stacks and stronger spectral capability.

Cislunar / Lunar-Orbit Instruments

Instrument	Type	Energy Range	Platform	Status
CRaTER	Si pairs + tissue-equivalent plastic	ND	LRO lunar orbit	flown
RADOM	Mini dosimeter spectrometer	ND	Chandrayaan-1 lunar orbit	flown
MATCH	DSSD + BGO	MeV range	Chang'E-7	development
PixPAN	Magnetic spectrometer + Timepix4	0.1–10 GeV	ESA LunPAN	concept
NeuPix	Hybrid pixel detector	10–100 MeV	ESA LunPAN	concept
IRADCAL	Scintillator + SiPM	keV–100 MeV	2U CubeSat concept	concept

Cislunar developments are increasingly hybrid, with explicit focus on lunar albedo particles, neutrons and gamma-rays.

Comparison Across Orbit Regimes

Orbit Regime	Dominant Detector Style	Main Driver	Typical
LEO	Pixel / compact Si / dosimetry	mapping, dose, variability	low SV
MEO	Silicon telescope stacks	trapped belts / spectra	higher f
Cislunar	Hybrid pixel + scintillator + spectrometer	SEP, GCR, albedo, neutrons	multi-species

- LEO: compact deployment and distributed monitoring
- MEO: spectral discrimination in radiation belts
- Cislunar: broad environment, secondaries from regolith, neutron sensitivity

Physics Implications

- **LEO:** compact instruments are often sufficient for dose monitoring and spatial mapping
- **MEO:** thicker detector stacks are required to resolve trapped particle populations
- **Cislunar:** instrument requirements broaden to include:
  - SEP protons in the 10–100 MeV range
  - GCR response at higher energies
  - secondary neutrons and gamma-rays from regolith
- This is why recent cislunar concepts are increasingly hybrid rather than pure silicon telescope or pure pixel systems

Summary

- Grouping instruments by orbit clarifies both technology choice and physics driver
- Red entries identify instruments absent from the original baseline list
- Blue entries identify baseline instruments whose status / performance has changed
- LEO remains dominated by compact and pixel-based monitors
- MEO remains the domain of silicon telescope systems
- Cislunar developments show the clearest shift toward hybrid architectures